

What is the central idea in Cell structure?

Light microscopes resolve cells and tissues; electron microscopes resolve organelles and membranes.  
Magnification = image size/actual size, with matching units.

9700-01-C1

QUESTION + ANSWER

A micrograph shows a 12 mm chloroplast that is actually 6  $\mu\text{m}$  long. Calculate magnification and name one organelle present in plant cells but absent from typical animal cells.

12 mm = 12 000  $\mu\text{m}$ ; magnification = 2000. Chloroplast or large permanent vacuole.

9700-01-P

QUESTION + ANSWER

What bond joins amino acids in a protein?

- A Glycosidic
- B Peptide
- C Ester
- D Hydrogen only

B. Peptide

Why: Monomers condense into polymers, releasing water; hydrolysis breaks bonds with water.

9700-02-M

QUESTION + ANSWER

What happens to a competitive inhibitor's effect when substrate increases strongly?

- A It always becomes stronger
- B The enzyme changes into product
- C It may be overcome
- D Temperature becomes irrelevant

C. It may be overcome

Why: Enzymes lower activation energy by binding substrate at an active site and stabilising a transition state.

9700-03-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Cell structure?

Prokaryotes lack a membrane-bound nucleus and 80S ribosomes; eukaryotic organelles compartmentalise metabolic reactions. Resolution, unlike magnification, determines the smallest separable detail.

9700-01-C2

QUESTION + ANSWER

What is the central idea in Biological molecules?

Monomers condense into polymers, releasing water; hydrolysis breaks bonds with water. Carbohydrates have glycosidic bonds, proteins peptide bonds and nucleic acids phosphodiester bonds.

9700-02-C1

QUESTION + ANSWER

How many water molecules form when one chain of seven amino acids assembles from free amino acids? Explain.

Six peptide bonds are formed, so six condensation events produce six water molecules.

9700-02-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Enzymes?

Temperature increases collision rates until heat disrupts protein structure; pH can change ionisation of active-site groups. Competitive inhibitors increase apparent substrate needed; non-competitive inhibitors lower maximum rate.

9700-03-C2

QUESTION + ANSWER

What determines the smallest resolvable detail in a micrograph?

- A Resolution
- B Image colour
- C Printed page size
- D Number of labels

A. Resolution

Why: Light microscopes resolve cells and tissues; electron microscopes resolve organelles and membranes.

9700-01-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Biological molecules?

The sequence of amino acids determines how a protein folds and thus its specificity; denaturation disrupts higher structure without usually hydrolysing peptide bonds.

9700-02-C2

QUESTION + ANSWER

What is the central idea in Enzymes?

Enzymes lower activation energy by binding substrate at an active site and stabilising a transition state. Initial rate is a fair comparison because substrate depletion and product accumulation are smallest early on.

9700-03-C1

QUESTION + ANSWER

Two enzymes have equal maximum activity, but enzyme X needs higher substrate concentration to reach half-maximal rate. Give one possible explanation and how to test it.

Competitive inhibition of X is possible; repeat with increasing substrate concentrations; if inhibition can be overcome at high substrate, both approach similar maximum rate.

9700-03-P

QUESTION + ANSWER

FagusED · A-Level 9700  
4 Cell membranes and transport

What is the central idea in Cell membranes and transport?

A phospholipid bilayer has hydrophilic heads facing water and hydrophobic tails inward. Proteins support facilitated diffusion, active transport and signalling.

9700-04-C1

QUESTION + ANSWER

FagusED · A-Level 9700  
4 Cell membranes and transport

Explain why a plant cell loses turgor in concentrated sucrose but may stay viable.

Outside solution has lower water potential; water exits by osmosis through the partially permeable membrane; membrane pulls from wall (plasmolysis), but wall remains intact and cell can recover.

9700-04-P

QUESTION + ANSWER

FagusED · A-Level 9700  
5 The mitotic cell cycle

What separates to opposite poles during mitosis?

- A Sister chromatids
- B Homologous chromosomes only
- C Cell walls
- D Two intact nuclei

A. Sister chromatids  
Why: Interphase includes DNA replication before mitosis.

9700-05-M

QUESTION + ANSWER

FagusED · A-Level 9700  
6 Nucleic acids and protein synthesis

Which mechanism, condition or distinction matters in Nucleic acids and protein synthesis?

Transcription produces RNA from a DNA template; translation reads mRNA codons at ribosomes while tRNA brings amino acids. A substitution may be silent, missense or nonsense depending on codon and reading frame.

9700-06-C2

QUESTION + ANSWER

FagusED · A-Level 9700  
4 Cell membranes and transport

Which mechanism, condition or distinction matters in Cell membranes and transport?

Diffusion proceeds down a concentration gradient; osmosis is net water movement from higher to lower water potential through a partially permeable membrane. Active transport uses ATP to move substances against gradients.

9700-04-C2

QUESTION + ANSWER

FagusED · A-Level 9700  
5 The mitotic cell cycle

What is the central idea in The mitotic cell cycle?

Interphase includes DNA replication before mitosis. Mitosis separates identical sister chromatids so daughter nuclei retain chromosome number and genetic identity, subject to mutation.

9700-05-C1

QUESTION + ANSWER

FagusED · A-Level 9700  
5 The mitotic cell cycle

A field has 18 dividing cells among 150 cells. Calculate mitotic index and explain why daughter cells have equal chromosome number.

Index =  $18/150 = 0.12$ . DNA replicated in interphase, then sister chromatids separate to opposite poles.

9700-05-P

QUESTION + ANSWER

FagusED · A-Level 9700  
6 Nucleic acids and protein synthesis

A DNA template segment 3'-TAC GAA-5' is transcribed. Give mRNA 5'-3' and explain why changing one base does not always change protein.

mRNA 5'-AUG CUU-3'. Genetic code is degenerate: a changed codon can specify the same amino acid.

9700-06-P

QUESTION + ANSWER

FagusED · A-Level 9700  
4 Cell membranes and transport

What directly drives active transport against a gradient?

- A Osmosis
- B Simple diffusion
- C Gravity
- D ATP use

D. ATP use  
Why: A phospholipid bilayer has hydrophilic heads facing water and hydrophobic tails inward.

9700-04-M

QUESTION + ANSWER

FagusED · A-Level 9700  
5 The mitotic cell cycle

Which mechanism, condition or distinction matters in The mitotic cell cycle?

Cell cycle checkpoints restrict division after DNA damage; uncontrolled proliferation can lead to tumours. A mitotic index estimates the fraction of observed cells undergoing mitosis.

9700-05-C2

QUESTION + ANSWER

FagusED · A-Level 9700  
6 Nucleic acids and protein synthesis

What is the central idea in Nucleic acids and protein synthesis?

DNA is antiparallel, with A-T and C-G complementary pairing. Semi-conservative replication yields one old and one newly synthesised strand per molecule.

9700-06-C1

QUESTION + ANSWER

FagusED · A-Level 9700  
6 Nucleic acids and protein synthesis

Which molecule carries codons read by ribosomes?

- A DNA polymerase
- B mRNA
- C ATP synthase
- D Haemoglobin

B. mRNA  
Why: DNA is antiparallel, with A-T and C-G complementary pairing.

9700-06-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**7 Transport in plants**

What is the central idea in Transport in plants?

Xylem dead lignified vessels carry water upward under transpiration pull; cohesion and adhesion maintain water columns.

9700-07-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**7 Transport in plants**

Describe how increasing wind speed may change transpiration rate and why the effect can plateau.

Wind removes humid boundary layer, increasing water-vapour gradient and evaporation; rate may become limited by stomatal aperture/water supply or closure in water stress.

9700-07-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**8 Transport in mammals**

What is the Bohr effect useful for in active tissue?

- A More blood clotting
- B Lower CO<sub>2</sub>
- C No diffusion
- D Oxygen unloading

D. Oxygen unloading  
Why: Double circulation sends blood through pulmonary and systemic circuits; separation permits high systemic pressure and lower pulmonary pressure.

9700-08-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**9 Gas exchange**

What adaptation lowers diffusion distance at alveoli?

- A Thin epithelium
- B Thick cartilage
- C Small total area
- D No blood supply

A. Thin epithelium  
Why: Ventilation refreshes alveolar air and maintains oxygen and carbon dioxide partial-pressure gradients.

9700-09-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**7 Transport in plants**

Which mechanism, condition or distinction matters in Transport in plants?

Phloem living sieve tubes translocate sucrose from sources to sinks, driven by loading, osmotic water movement and pressure flow. Water potential differences at root, leaf and atmosphere sustain net transport.

9700-07-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**8 Transport in mammals**

What is the central idea in Transport in mammals?

Double circulation sends blood through pulmonary and systemic circuits; separation permits high systemic pressure and lower pulmonary pressure. Arteries have thick elastic walls, veins valves and larger lumens; capillaries have thin walls for exchange.

9700-08-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**8 Transport in mammals**

Explain two structural adaptations of a capillary for exchange and why oxygen unloading is greater in active tissues.

Thin endothelial wall minimises diffusion distance; narrow lumen/slower flow or extensive network increases exchange time/surface; high CO<sub>2</sub> lowers haemoglobin affinity (Bohr effect).

9700-08-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**9 Gas exchange**

Explain why pulmonary fibrosis reduces oxygen transfer even if alveolar oxygen partial pressure stays constant.

Fibrosis thickens diffusion barrier; diffusion distance increases, slowing oxygen uptake into blood.

9700-09-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**7 Transport in plants**

Which tissue translocates sucrose from source to sink?

- A Xylem vessel
- B Epidermis
- C Phloem sieve tubes
- D Palisade cells

C. Phloem sieve tubes  
Why: Xylem dead lignified vessels carry water upward under transpiration pull; cohesion and adhesion maintain water columns.

9700-07-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**8 Transport in mammals**

Which mechanism, condition or distinction matters in Transport in mammals?

Haemoglobin's cooperative oxygen binding produces a sigmoidal dissociation curve; a lower affinity in metabolically active tissue assists unloading.

9700-08-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**9 Gas exchange**

What is the central idea in Gas exchange?

Ventilation refreshes alveolar air and maintains oxygen and carbon dioxide partial-pressure gradients. Numerous alveoli provide a large surface, a thin epithelium short diffusion path and capillary networks carry gases away.

9700-09-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**9 Gas exchange**

Which mechanism, condition or distinction matters in Gas exchange?

Insects have tracheae and tracheoles delivering air toward cells, while stomata regulate plant gas exchange.

9700-09-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**10 Infectious diseases**

What is the central idea in Infectious diseases?

A pathogen causes transmissible disease; viruses replicate inside host cells, while bacteria may divide independently. Transmission can involve droplets, water or vectors, so prevention must target the route.

9700-10-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**10 Infectious diseases**

Which mechanism, condition or distinction matters in Infectious diseases?

Antibiotics act on bacterial structures/processes but do not treat viruses. Resistance evolves when a heritable variant survives exposure and reproduces.

9700-10-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**10 Infectious diseases**

Why do antibiotics not directly treat viral replication?

A Viruses have more ribosomes  
B Viruses lack targeted bacterial processes  
C Antibiotics only work in air  
D All viruses are bacteria

B. Viruses lack targeted bacterial processes

Why: A pathogen causes transmissible disease; viruses replicate inside host cells, while bacteria may divide independently.

9700-10-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**10 Infectious diseases**

A bacterial population survives an antibiotic treatment. Explain selection of resistance in three linked steps.

Heritable resistance variants pre-exist or arise by mutation; antibiotic kills susceptible cells so resistant cells survive; resistant cells reproduce and allele frequency rises.

9700-10-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**11 Immunity**

What is the central idea in Immunity?

Phagocytes engulf pathogens non-specifically; lymphocytes respond to specific antigens. B cells differentiate into plasma cells producing antibodies and memory cells enabling faster secondary responses.

9700-11-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**11 Immunity**

Which mechanism, condition or distinction matters in Immunity?

T helper cells coordinate responses; T killer cells destroy infected cells. A vaccine supplies antigen safely to induce immune memory; it does not directly deliver permanent ready-made antibodies.

9700-11-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**11 Immunity**

Distinguish active vaccination from passive antibody transfer in onset and duration of protection.

Passive transfer supplies antibodies and acts immediately, but is temporary as antibodies degrade; active vaccination induces memory cells and lasts longer after a slower primary response.

9700-11-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**11 Immunity**

What enables a faster secondary antibody response?

A Erythrocytes  
B Platelets  
C Memory cells  
D Antigens disappear forever

C. Memory cells

Why: Phagocytes engulf pathogens non-specifically; lymphocytes respond to specific antigens.

9700-11-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**12 Energy and respiration**

What is the central idea in Energy and respiration?

Glycolysis in cytoplasm converts glucose to pyruvate with a small ATP yield; link reaction and Krebs cycle feed reduced coenzymes into electron transport.

9700-12-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**12 Energy and respiration**

Which mechanism, condition or distinction matters in Energy and respiration?

Oxidative phosphorylation uses a proton gradient across inner mitochondrial membrane to drive ATP synthase. Oxygen is final electron acceptor. Anaerobic pathways regenerate NAD so glycolysis can continue.

9700-12-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**12 Energy and respiration**

What is oxygen's role in aerobic respiration?

A First glycolysis substrate  
B Only carbon source  
C ATP synthase motor  
D Final electron acceptor

D. Final electron acceptor

Why: Glycolysis in cytoplasm converts glucose to pyruvate with a small ATP yield; link reaction and Krebs cycle feed reduced coenzymes into electron transport.

9700-12-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**12 Energy and respiration**

Explain why cyanide blocking electron transport rapidly lowers ATP production despite available glucose.

Electrons cannot flow to oxygen; proton gradient is not maintained, so ATP synthase makes far less ATP; only glycolysis remains.

9700-12-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**13 Photosynthesis**

What is the central idea in Photosynthesis?

Photosystems in thylakoids absorb light, split water and make ATP and reduced NADP. The Calvin cycle in the stroma uses ATP and reduced NADP to fix CO<sub>2</sub> into triose phosphate; RuBP is regenerated.

9700-13-C1

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**14 Homeostasis**

What is the central idea in Homeostasis?

Negative feedback detects deviation from a set point and produces a response opposing the change. Pancreatic  $\beta$  cells release insulin at high blood glucose, increasing uptake and glycogen synthesis;  $\alpha$  cells release glucagon at low glucose, promoting glycogen breakdown and gluconeogenesis.

9700-14-C1

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**15 Control and coordination**

What is the central idea in Control and coordination?

A nervous impulse follows depolarisation when voltage-gated sodium channels open and repolarisation via potassium channels. Myelin permits saltatory conduction.

9700-15-C1

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**15 Control and coordination**

Explain how myelination increases conduction speed and why chemical synapse transmission is directional.

Action potentials regenerate at nodes of Ranvier, reducing membrane to depolarise; transmitter vesicles are presynaptic and receptor proteins postsynaptic.

9700-15-P

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**13 Photosynthesis**

Which mechanism, condition or distinction matters in Photosynthesis?

Light intensity, CO<sub>2</sub> and temperature may each limit rate, with changing limiting factor as conditions change.

9700-13-C2

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**13 Photosynthesis**

A plant's oxygen production plateaus when light intensity doubles. State two possible limits and suggest one controlled test.

CO<sub>2</sub> availability or enzyme temperature may limit; vary CO<sub>2</sub> while holding light and temperature constant and observe whether oxygen production rises.

9700-13-P

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**14 Homeostasis**

Which hormone usually increases after blood glucose rises?

- A Glucagon
- B Insulin
- C ADH always
- D Adrenaline only

B. Insulin  
Why: Negative feedback detects deviation from a set point and produces a response opposing the change.

9700-14-M

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**15 Control and coordination**

Why can myelin increase impulse conduction speed?

- A It removes ion channels everywhere
- B It reverses impulse direction
- C Impulses regenerate at nodes
- D It makes synapses electrical

C. Impulses regenerate at nodes  
Why: A nervous impulse follows depolarisation when voltage-gated sodium channels open and repolarisation via potassium channels.

9700-15-M

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**13 Photosynthesis**

Where does Calvin cycle carbon fixation occur?

- A Stroma
- B Thylakoid lumen
- C Mitochondrial matrix
- D Cytoplasm only

A. Stroma  
Why: Photosystems in thylakoids absorb light, split water and make ATP and reduced NADP.

9700-13-M

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**14 Homeostasis**

Which mechanism, condition or distinction matters in Homeostasis?

Kidney nephrons filter blood and selectively reabsorb useful solutes; ADH increases water permeability of collecting ducts.

9700-14-C2

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**14 Homeostasis**

Explain the response of insulin and glucagon after a carbohydrate-rich meal.

Blood glucose increases; insulin increases and promotes cellular uptake/storage as glycogen; glucagon secretion decreases.

9700-14-P

[QUESTION + ANSWER](#)

**FagusED · A-Level 9700**  
**15 Control and coordination**

Which mechanism, condition or distinction matters in Control and coordination?

Chemical synapses release neurotransmitter following calcium influx; transmission is one way because vesicles and receptor sites differ across the cleft. Hormones travel in blood and generally act more slowly but for longer.

9700-15-C2

[QUESTION + ANSWER](#)

What is the central idea in Inheritance?

Meiosis produces haploid gametes and variation by crossing over and independent assortment. At fertilisation diploid number is restored.

9700-16-C1

QUESTION + ANSWER

Two heterozygous parents produce 80 offspring; 18 show a recessive phenotype. Calculate expected number under a 3:1 ratio and give one cause of deviation.

Expected recessive =  $80/4 = 20$ ; observed 18 can differ by sampling chance; mutation/selection or flawed model are other acceptable reasons.

9700-16-P

QUESTION + ANSWER

At the population level, what changes during evolution?

- A Allele frequencies
- B Each organism's inherited DNA by need
- C Species names only
- D Every trait equally

A. Allele frequencies  
Why: Mutation produces new alleles; natural selection changes allele frequencies when variants affect survival and reproductive success in a given environment.

9700-17-M

QUESTION + ANSWER

State two reasons genetic diversity matters to a threatened population and one limitation of ex situ conservation.

Different alleles improve capacity to respond to environmental change and reduce inbreeding effects; ex situ populations may be small/lose habitat behaviour or require reintroduction.

9700-18-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Inheritance?

For independently assorting genes, multiply probabilities; for linkage, recombinant classes are less frequent than predicted by independent assortment. A chi-square test compares observed and expected counts but does not prove the null hypothesis true.

9700-16-C2

QUESTION + ANSWER

What is the central idea in Selection and evolution?

Mutation produces new alleles; natural selection changes allele frequencies when variants affect survival and reproductive success in a given environment.

9700-17-C1

QUESTION + ANSWER

In a population exposed to a new drug, explain why resistance increases over generations without claiming individual organisms adapt by need.

Random inherited variants arise before/during exposure; resistant individuals survive and reproduce more; resistant allele frequency rises in the population.

9700-17-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Classification, biodiversity and conservation?

A high species richness alone does not describe evenness; diversity indices combine both. Conservation can protect habitats in situ and populations ex situ; maintain genetic variation to reduce vulnerability.

9700-18-C2

QUESTION + ANSWER

What can crossing over during meiosis produce?

- A Identical clones only
- B Doubled chromosome number always
- C Loss of all genes
- D New allele combinations

D. New allele combinations  
Why: Meiosis produces haploid gametes and variation by crossing over and independent assortment.

9700-16-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Selection and evolution?

Genetic drift can cause chance changes, especially in small populations. Speciation requires reproductive isolation followed by genetic divergence. Selection acts on phenotypes, while populations evolve across generations.

9700-17-C2

QUESTION + ANSWER

What is the central idea in Classification, biodiversity and conservation?

Classification reflects evolutionary relationships using shared traits and molecular sequence data. Biodiversity can be considered at genetic, species and ecosystem levels.

9700-18-C1

QUESTION + ANSWER

Which pair matters for species diversity?

- A Climate and pH only
- B Richness and evenness
- C Only total individual count
- D Only DNA molecule length

B. Richness and evenness  
Why: Classification reflects evolutionary relationships using shared traits and molecular sequence data.

9700-18-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**19 Genetic technology**

What is the central idea in Genetic technology?

PCR cycles denature DNA, anneal primers and extend strands with thermostable polymerase; product can double approximately each cycle under ideal conditions.

9700-19-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**19 Genetic technology**

A PCR sample starts with 10 double-stranded targets. Estimate ideal copies after six cycles and give one reason actual yield differs.

$10 \times 2^6 = 640$  copies; reagent depletion, incomplete efficiency or inhibitors reduce actual yield.

9700-19-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P3 Advanced practical skills**

Why use biological replicates in a practical?

A To change several conditions  
B To avoid labelling axes  
C To make one measurement exact  
D To estimate variation

D. To estimate variation  
Why: Biology practical work requires clear variable definitions, biological replicates and consistent measurement.

9700-20-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P5 Planning, analysis and evaluation**

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

Differentiate correlation from causation; choose a suitable graph/statistical test and interpret it against assumptions. Improvements should respond to limits such as uncontrolled temperature or small sample size.

9700-21-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**19 Genetic technology**

Which mechanism, condition or distinction matters in Genetic technology?

Restriction enzymes cut defined sequences, ligase joins DNA, and vectors introduce constructs into cells. Gel electrophoresis separates fragments by size; controls are needed to interpret band patterns and limit contamination.

9700-19-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P3 Advanced practical skills**

What is the central idea in P3 Advanced practical skills?

Biology practical work requires clear variable definitions, biological replicates and consistent measurement.

9700-20-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P3 Advanced practical skills**

A student studies sucrose concentration and potato mass change. Identify independent, dependent and two controlled variables.

Independent: sucrose concentration; dependent: change or percentage change in potato mass; controls: piece surface/starting mass, immersion time, temperature or solution volume.

9700-20-P

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P5 Planning, analysis and evaluation**

What would strengthen a proposed causal explanation?

A Controlled intervention and repeats  
B A larger graph title  
C One unreplicated observation  
D Removing units

A. Controlled intervention and repeats  
Why: Start with a biological hypothesis and testable prediction, then design sampling to reduce bias and enough repeats to estimate variability.

9700-21-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**19 Genetic technology**

Which enzyme extends primers in PCR?

A DNA ligase  
B Restriction endonuclease  
C Thermostable DNA polymerase  
D RNA polymerase only

C. Thermostable DNA polymerase  
Why: PCR cycles denature DNA, anneal primers and extend strands with thermostable polymerase; product can double approximately each cycle under ideal conditions.

9700-19-M

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P3 Advanced practical skills**

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

Use a microscope scale calibrated by stage micrometer; in enzyme or osmosis studies hold temperature, pH and initial material comparable. Present calculated rates with units and graphs with suitably labelled axes.

9700-20-C2

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P5 Planning, analysis and evaluation**

What is the central idea in P5 Planning, analysis and evaluation?

Start with a biological hypothesis and testable prediction, then design sampling to reduce bias and enough repeats to estimate variability.

9700-21-C1

QUESTION + ANSWER

**FagusED · A-Level 9700**  
**P5 Planning, analysis and evaluation**

A study finds leaf area and transpiration rate correlated. Explain two reasons this does not alone establish causation and propose one improvement.

Other factors such as stomatal density or temperature may covary; correlation gives association not controlled intervention; hold conditions constant/normalise per unit area and replicate.

9700-21-P

QUESTION + ANSWER